Tianxing Zhu, PhD

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Technical leader with 12+ years of R&D and engineering experience, including 6+ years of team leadership, now focused on XR optics, human visual perception, and applied AI. Built Instrument Systems' Shanghai XR optical laboratory and engineering team, and now lead XR R&D in China: defining near-eye display measurement methods, translating them into production tests, and contributing to international standards development. Led technical engagements with 20+ device makers and display manufacturers, including Apple, Samsung, and BOE. Lead author of the *XR Optical Testing Handbook*, with 20+ peer-reviewed papers and 7+ invited XR talks.

TECHNICAL EXPERTISE

- **XR Systems & Perception:** Near-Eye Display Systems, Perceptual Image Quality Evaluation, Display Performance Optimization (Android XR, OpenCV).
- **Agentic AI & Autonomous Experimentation:** Wearable XR+AI Systems, AI-Driven Optical Harnesses, Agent Orchestration, Multimodal AI Integration (Python, PyTorch).
- **Optical Metrology & Computational Imaging:** Optical Characterization Systems, Optical & Colorimetric Modeling, Display Calibration, Production Test & Validation, Hyperspectral Imaging (C#, Optical Metrology & Instrument Control Suites).

PROFESSIONAL EXPERIENCE

Instrument Systems · **Konica Minolta Group** — Shanghai, China

Technical Manager — *2025–Present*

- **XR Technical Strategy & Industry Collaboration:** Lead XR technical research and development in China, defining strategic priorities and collaborating with global R&D teams, device manufacturers, and optical suppliers to shape evaluation methodologies, technical roadmaps, and product requirements for next-generation near-eye display systems. Drive ongoing international standards development and serve as lead author of the *XR Optical Testing Handbook*.
- **XR Systems Architecture:** Architected an application-driven XR systems framework translating smart-glasses use cases and human visual perception into optical specifications, measurement methodologies, and platform standards. Bridged dynamic eye movement with static lab metrics, resolved spatial alignment and optical distortion across virtual imagery, real-world scenes, and prescription lenses, and distilled complex lab characterizations into rapid production tests.
- **AI-Driven Optical Harness Platforms:** Designed and developed a goal-driven, self-recursive hardware-in-the-loop (HIL) AI research platform integrating XR devices, optical instruments, and engineering software, where AI agents use physical feedback to evolve experimental methods, control software, and computational models toward defined research objectives.
- **Perceptual Evaluation & Optical Optimization:** Lead human-centric evaluation and optimization of see-through near-eye display quality under real-world viewing conditions. Proposed a shared-gamut color space and inverse rendering framework for additive AR, modeled spectral filtering and color prediction across diffractive and geometric waveguides, and optimized perceptual contrast and spatial uniformity across varying ambient light.

Application Manager — *2020–2024*

- **XR Laboratory & Team Development:** Founded Instrument Systems' Shanghai XR optical laboratory in 2024 and built the local engineering team, establishing full-field, spectral, and angular characterization capabilities for near-eye display evaluation and customer programs.
- **Optical Systems Integration:** Led technical engagements with 20+ global device makers and display manufacturers, including Apple, Samsung, and BOE, translating complex optical measurement requirements into customized testing solutions, software integrations, and production-ready systems from proof of concept through deployment.
- **Optical Automation & Applied AI:** Delivered 12+ optical automation projects across China, Vietnam, and Singapore, streamlining instrument calibration, certificate generation, and testing workflows. Developed AI-based display mura detection and an AI-assisted knowledge system for technical troubleshooting and training.

Instrument Systems · Konica Minolta Group Shanghai, China

Senior Application Engineer *2017–2019*

- **Application Engineering:** Established regional application engineering and local calibration capabilities, providing technical consultation, system integration, diagnostics, deployment, and training for optical measurement systems.
- **Calibration Process Automation:** Developed an end-to-end spectrometer calibration platform, increasing daily throughput by 167% and reducing 100+ manual operations to a three-step workflow, standardizing calibration and certificate generation across China and Vietnam.

Saint-Gobain Research Center Shanghai, China

Research Engineer *2014–2017*

- **Product Upgrades:** Delivered two major product-line upgrades for Saint-Gobain Abrasives through engineering development, validation, and industrialization, contributing 40% of the business unit's 2016 revenue.
- **Team Leadership:** Led a four-person engineering team, guiding physics-based research and technical decision-making to balance product performance, global design specifications, Asian manufacturing capabilities, and customer requirements.

STANDARDS & PROFESSIONAL SERVICE

International Committee for Display Metrology (ICDM)
Voting Expert Member · 2025–Present · Contribute to international display metrology methods and standards through the Near-Eye Display and Temporal Measurements subcommittees.

Society for Information Display (SID) China
Technical Committee Member · 2025–Present · Serve on the VR & AR and Display Measurement subcommittees.

SELECTED INVITED TALKS

Delivered 7+ invited XR presentations at international conferences and exhibitions since 2023.

EPIC TechWatch at China International Optoelectronic Exposition (CIOE 2026)
Shenzhen · September 9, 2026 · "Spectral Trade-Offs in Diffractive AR Waveguides."

SID Symposium on New Display Technologies
Qingdao · August 31, 2026 · "Advances in XR Visual Perception Quality and Evaluation."

International Conference on Display Technology (ICDT)
2025–2026 · Delivered four presentations on AR visual quality, optical metrology, and near-eye display evaluation.

SELECTED PUBLICATIONS & RESEARCH

20+ peer-reviewed papers. Full publication list: tzhu64.github.io/publications

XR Optical Testing Handbook
T. Zhu and M. Hillebrand · 2026 · ISBN: 978-7-89569-660-0

Shared-Gamut Color Space and Inverse Rendering for Additive Optical See-Through Displays
T. Zhu et al. · Submitted to *SID Display Week*, 2027

Goal-Driven Autonomous Optical Experimentation via a Self-Recursive Hardware–Software Harness
T. Zhu et al. · Submitted to *SID Display Week*, 2027

Spatial Memory for AR Glasses: Hierarchical Visual Acquisition and Source-Linked Recall
T. Zhu and D. Huang · Submitted to *ICDT*, 2027

Gaze vs. View: A Framework for Correlating Dynamic and Static Measurements in AR Optics
T. Zhu et al. · *ICDT*, 2026

The Impact of Waveguide Spectral Filtering on AR Color Fidelity
T. Zhu et al. · *ICDT*, 2026

EDUCATION

PhD, Software Engineering (AI) Tongji University 2020–2025

MS, Mechanical Engineering Tsinghua University 2011–2014

BS, Mechanical Engineering University of Toronto 2006–2010